

Tensor computations

Comon's exact-rank conjecture for Hankel tensors

Difficulty: challenging **Importance:** interesting to the community

Statement

For every $m \geq 3$, $n \geq 2$, and $h \in \mathbb{C}^{m(n-1)+1}$, form the symmetric tensor

$$H_{i_1 \dots i_m} = h_{i_1 + \dots + i_m - m}, \quad 1 \leq i_j \leq n.$$

Is $R(H) = R_{\text{sym}}(H)$ always true? Here $R(H)$ is the least integer $r \geq 0$ admitting

$$H = \sum_{j=1}^r u_{j,1} \otimes \dots \otimes u_{j,m}, \quad u_{j,k} \in \mathbb{C}^n,$$

whereas $R_{\text{sym}}(H)$ is the least r admitting $H = \sum_{j=1}^r c_j v_j^{\otimes m}$, with $c_j \in \mathbb{C}$ and $v_j \in \mathbb{C}^n$. Empty sums represent zero. No genericity or Vandermonde restriction is imposed.

Relevance

This would justify symmetry-preserving decomposition lengths for every Hankel tensor, including exceptional data. Such structure appears in harmonic retrieval and structured tensor factorization.

References

1. J. Nie and K. Ye, *Hankel Tensor Decompositions and Ranks*, SIAM J. Matrix Anal. Appl. 40 (2019). DOI; [primary preprint](#), §6, Conjecture 6.3, p.16.
2. L. Qi, *Hankel Tensors: Associated Hankel Matrices and Vandermonde Decomposition*, 2014. [Primary preprint](#), §1, equation (1), and §4, Theorem 3, for Hankel structure and decomposition.

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Nie and Ye formulate this restricted conjecture after explicitly acknowledging Shitov's counterexample to the unrestricted symmetric-tensor conjecture. Searches "Hankel" "Conjecture 6.3", "Hankel" "Comon" "counterexample", and "Hankel tensors" ranks conjectures solved located no solution. No later primary reaffirmation was located; the dated conclusion is bounded by this search.